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ML MODEL TO DETECT INTRACRANIAL PRESSURE

Submitted in partial fulfilment of the requirements as a part of the curriculum,
Bachelors of Engineering in Artificial Intelligence and Machine Learning

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CERTIFICATE

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Ms. S Santhiya

(Dr. Shyam P Joy)

Signature of the Guide

Signature of the HOD

DECLARATION

We, the undersigned, students of **CMR Institute of Technology**, hereby declare that this mini project report titled

“ML Model to Detect Intracranial Pressure”

is a record of our original work done under the supervision of ***Ms. S Santhiya, Assistant Professor***, Department of ***Artificial Intelligence and Machine Learning***. And has not formed the basis of award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

Intracranial Pressure (ICP) monitoring plays an important role in diagnosing serious neurological conditions such as traumatic brain injury, brain tumours, and internal bleeding. Traditional ICP monitoring methods are generally invasive, expensive, and require specialized medical procedures, which may increase patient risk and limit accessibility. This project presents a Machine Learning-based system for detecting abnormal intracranial pressure conditions using medical datasets and predictive algorithms.

The proposed system was developed using Python, Scikit-learn, Flask, and React technologies. The dataset was pre-processed using techniques such as missing value handling, duplicate removal, normalization, and feature selection to improve prediction accuracy. Multiple machine learning algorithms including Logistic Regression, Decision Tree, Support Vector Machine, K-Nearest Neighbours, and Random Forest were tested and compared. Among them, the Random Forest Classifier achieved the highest prediction accuracy.

The developed system provides a user-friendly interface for entering patient data and generating prediction results. Experimental results demonstrate that the proposed model can effectively predict ICP conditions with good accuracy and reliability. The project highlights the potential of machine learning in healthcare applications and supports early diagnosis and decision-making for medical professionals.

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1. INTRODUCTION

1.1 Overview

In recent years, the healthcare industry has witnessed rapid advancements in Artificial Intelligence (AI) and Machine Learning (ML). These technologies are being increasingly used for disease diagnosis, medical imaging, patient monitoring, and predictive healthcare systems. Machine learning enables computers to analyze medical data, identify hidden patterns, and assist doctors in making faster and more accurate clinical decisions. The integration of ML in healthcare has improved the efficiency and reliability of medical diagnosis while reducing human error.

Intracranial Pressure (ICP) refers to the pressure inside the skull caused by brain tissue, blood, and cerebrospinal fluid. Abnormal ICP levels can indicate serious neurological conditions such as traumatic brain injury, brain tumors, stroke, hydrocephalus, or internal bleeding. Continuous monitoring of ICP is extremely important because delayed detection may lead to severe brain damage or even death. Traditional ICP monitoring methods are often invasive, expensive, and require surgical procedures, which may increase the risk of infection and complications for patients.

The proposed project, “ML Model to Detect Intracranial Pressure,” focuses on developing a machine learning-based system capable of predicting or detecting abnormal intracranial pressure using medical parameters and patient data. The project aims to provide a faster, safer, and more efficient method for ICP analysis. By applying machine learning algorithms, the system can support healthcare professionals in early diagnosis and timely treatment, thereby improving patient outcomes and reducing medical risks.

1.2 Problem Statement

Intracranial Pressure monitoring is a critical process in diagnosing neurological disorders and brain-related injuries. Existing traditional methods for ICP monitoring are mostly invasive, costly, and require specialized medical equipment and surgical procedures. These methods may cause discomfort to patients and increase the chances of complications such as infection and bleeding.

Many hospitals and healthcare centres may not have access to advanced ICP monitoring systems, making early detection difficult. Delayed or inaccurate detection of abnormal intracranial pressure can result in severe health complications, permanent brain damage, or death. Therefore, there is a need for an intelligent, accurate, and non-invasive system that can assist healthcare professionals in detecting intracranial pressure effectively using machine learning techniques.

1.3 Objectives

1. To develop a machine learning model for detecting abnormal intracranial pressure levels.
2. To collect and preprocess medical data required for ICP prediction.
3. To implement suitable machine learning algorithms for accurate analysis and prediction.
4. To evaluate the performance of the developed model using accuracy and other evaluation metrics.
5. To provide a simple and efficient system that assists healthcare professionals in early diagnosis.
6. To reduce dependency on invasive ICP monitoring methods through intelligent prediction techniques.

1.4 Scope of the Project

This project focuses on designing and developing a machine learning-based system for detecting intracranial pressure using available medical datasets and patient-related parameters. The system includes data preprocessing, feature analysis, model training, prediction, and performance evaluation. The project aims to assist healthcare professionals by providing an intelligent tool for early ICP detection and analysis.

The project is limited to software-based prediction and analysis using machine learning techniques. It does not include real-time hardware implementation, surgical monitoring devices, or direct clinical diagnosis. The system is intended for educational and research purposes and should not replace professional medical judgment or advanced hospital equipment.

2. LITERATURE SURVEY

Several research works and studies have been carried out in the field of intracranial pressure monitoring using machine learning and signal processing techniques. The following literature survey highlights some important existing approaches, their contributions, and the limitations that motivated the development of the proposed project.

1. Machine Learning Approaches to Intracranial Pressure Prediction in Patients with Traumatic Brain Injury

This study presented a systematic review of machine learning methods used for predicting intracranial pressure crises in traumatic brain injury patients. The research discussed different

ML algorithms and their effectiveness in early ICP prediction using patient monitoring data. The study emphasized the importance of predictive systems in reducing secondary brain injuries. However, the research mainly focused on critical care environments and did not provide a simplified implementation suitable for educational or small-scale healthcare applications.

2. A Machine Learning Approach in the Non-Invasive Prediction of Intracranial Pressure Using Modified Photoplethysmography

This research proposed a non-invasive method for predicting intracranial pressure using photoplethysmography signals and machine learning algorithms. The system analyzed physiological signals and demonstrated that ML techniques can estimate ICP without invasive surgical procedures. The study achieved promising prediction accuracy and highlighted the potential of non-invasive monitoring methods. However, the approach required specialized signal acquisition techniques and advanced medical equipment, which may limit practical implementation in general healthcare systems.

3. Features Derived from Blood Pressure and Intracranial Pressure Predict Elevated ICP Events in Critically Ill Children

This paper focused on identifying elevated intracranial pressure events in critically ill children using machine learning techniques. Various physiological features such as blood pressure and ICP-related parameters were analysed using multiple ML models. The study showed that machine learning can successfully identify patterns associated with abnormal ICP events. Despite its effectiveness, the research was limited to paediatric critical care cases and relied heavily on continuously monitored clinical data.

4. A Supervised, Externally Validated Machine Learning Model for Artifact and Drainage Detection in High-Resolution ICP Monitoring Data

This research developed a supervised machine learning model to classify normal signals, artifacts, and drainage events in intracranial pressure monitoring data. The proposed model achieved high prediction accuracy and improved the reliability of ICP signal analysis. The study demonstrated the usefulness of ML in handling complex neurocritical care data. However, the system mainly focused on signal classification and artifact detection rather than providing a simplified ICP prediction model for broader healthcare usage.

5. Robust Peak Recognition in Intracranial Pressure Signals

This study introduced techniques for detecting and analysing waveform peaks in intracranial pressure signals. The research aimed to improve the monitoring and interpretation of ICP waveforms for neurological assessment. Signal processing and pattern recognition methods were used to enhance waveform analysis accuracy. Although the study contributed significantly to ICP signal analysis, it concentrated more on waveform processing rather than predictive machine learning-based diagnosis.

Summary of Literature Survey

From the reviewed literature, it is observed that machine learning techniques have significant potential in intracranial pressure monitoring and prediction. Existing systems mainly focus on complex hospital-based monitoring environments, advanced physiological signal analysis, or specialized medical equipment. The proposed project aims to develop a simpler and efficient machine learning-based ICP detection system that can assist healthcare professionals in early diagnosis and prediction using available medical datasets and ML algorithms.

3. METHODOLOGY

3.1 SYSTEM DESIGN

The proposed system is designed to detect and predict abnormal Intracranial Pressure (ICP) using Machine Learning techniques. The system follows a structured workflow that includes data collection, preprocessing, model training, prediction, and result generation. Medical data is first collected and cleaned to remove inconsistencies and missing values. The processed data is then provided to the machine learning model for training and prediction.

The system architecture is designed in a modular manner to ensure simplicity, scalability, and ease of implementation. Each module performs a specific function such as handling input data, preprocessing, prediction, and displaying results. The design allows efficient communication between modules and ensures reliable prediction of ICP conditions.

3.1.1 System Requirements

Functional Requirements

1. The system shall allow users to input patient medical data for analysis.
2. The system shall preprocess the input data by handling missing values and normalizing data.
3. The system shall train the machine learning model using the provided dataset.
4. The system shall predict whether the intracranial pressure level is normal or abnormal.
5. The system shall display prediction results clearly to the user.
6. The system shall store trained model data for future predictions.
7. The user should be able to view the prediction accuracy and performance metrics of the model.

Non-Functional Requirements

1. The system should provide accurate and reliable prediction results.
2. The system should have a simple and user-friendly interface.
3. The system should process data efficiently with minimum response time.
4. The system should maintain confidentiality and security of medical data.
5. The system should be scalable for handling larger datasets in the future.

6. The system should be maintainable and easy to update with new machine learning models.

Hardware and Software Requirements

Category	Development
OS	Windows 10 / Ubuntu 22.04
Processor	Intel Core i5 or above
GPU	Not Applicable / Optional
RAM	Minimum 8 GB RAM
Software	Python 3.x, VS Code, Kaggle / Medical Dataset
Libraries	Kaggle / Medical Dataset

3.1.2 System Architecture / High-Level Design (HLD)

The system architecture consists of multiple layers that work together to perform intracranial pressure prediction using machine learning techniques.

Layers in the System Architecture

1. Input Layer

- Collects patient medical parameters and dataset information.
- Accepts input data required for prediction.

2. Data Preprocessing Layer

- Cleans the dataset by removing missing or inconsistent values.
- Performs normalization and feature selection for better model performance.

3. Machine Learning Layer

- Trains the machine learning model using pre-processed data.
- Applies classification algorithms to predict ICP conditions.

4. Prediction Layer

- Generates prediction results based on trained model output.

- Classifies ICP as normal or abnormal.

5. Output Layer

- Displays prediction results and model accuracy.
- Provides final analysis to the user.

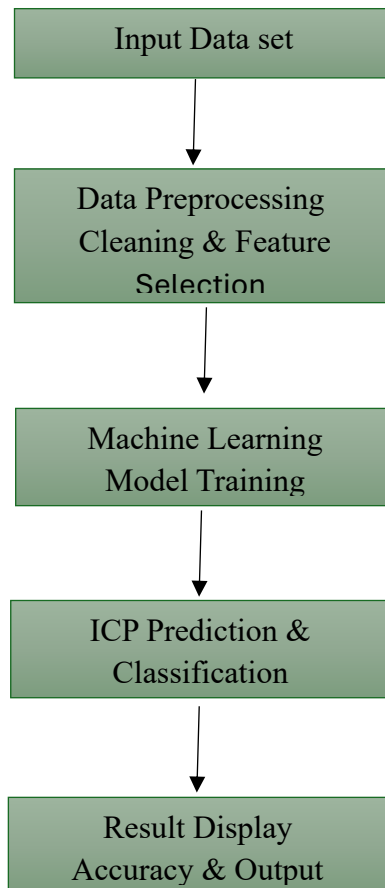


Figure 3.1: System Architecture Diagram

3.1.3 Module Description

No.	Module Name	Description
1	Data Collection Module	This module is responsible for collecting medical datasets related to intracranial pressure. The collected data is used as input for preprocessing and model training.
2	Data Preprocessing Module	This module cleans the dataset by handling missing values, normalization, and feature selection. It improves the quality of data before model training.

3	Machine Learning Model Module	This module trains and tests machine learning algorithms using the processed dataset. It performs ICP prediction based on learned patterns.
4	Prediction Module	This module accepts input parameters and predicts whether the intracranial pressure is normal or abnormal using the trained model.
5	Result Visualization Module	This module displays prediction results, accuracy scores, graphs, and evaluation metrics in a user-friendly format

3.1.4 Dataset Description

The dataset used in this project is a medical dataset related to intracranial pressure prediction collected from publicly available healthcare sources such as Kaggle and research publications. The dataset contains physiological and medical parameters associated with brain pressure monitoring.

Dataset Details

- **Source:** Kaggle / Public Medical Dataset
- **Type of Data:** Numerical medical data
- **Number of Samples:** Approximately 100000+ records
- **Number of Features:** Multiple physiological parameters
- **Features Include:** Blood pressure, heart rate, oxygen level, cerebral measurements, and related medical parameters
- **Target Variable:** Intracranial Pressure Condition (Normal / Abnormal)

Dataset Characteristics

- The dataset contains structured medical data.
- Missing values are handled during preprocessing.
- Data normalization is applied to improve model performance.
- The dataset is divided into training and testing sets for evaluation.

3.1.5 Models Used

Several machine learning algorithms were studied and evaluated for ICP prediction. Different classification models were compared based on accuracy, efficiency, and prediction performance.

Algorithms Considered

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier
4. Support Vector Machine (SVM)
5. K-Nearest Neighbours (KNN)

Final Model Chosen

The **Random Forest Classifier** was selected as the final model because it provided better prediction accuracy and handled medical data efficiently. Random Forest reduces overfitting by combining multiple decision trees and improves overall classification performance.

Hyperparameters Used

- Number of Estimators: 100
- Maximum Depth: 10
- Criterion: Gini Index
- Random State: 42

3.2 IMPLEMENTATION

This chapter explains the implementation details of the proposed “ML Model to Detect Intracranial Pressure” system. The implementation includes dataset handling, preprocessing, machine learning model development, prediction generation, and result visualization. Python-based machine learning libraries were used to build and evaluate the system due to their simplicity, efficiency, and strong support for data analysis applications.

3.2.1 Technology Stack

Layer	Technology / Tool	Reason for Choice
Frontend	React.js	Provides a fast, responsive, and component-based user interface for displaying prediction results dynamically
Backend	Python	Easy to implement machine learning applications and supports multiple ML libraries
Database	CSV File Storage	Lightweight and suitable for storing structured medical data
ML/AI Library	Scikit-learn	Provides efficient machine learning algorithms and easy-to-use interfaces
Deployment	Flask	Enables deployment of the trained ML model as a web application

3.2.2 Implementation of Key Modules

Module 1: Data Collection Module

The Data Collection Module is responsible for importing and managing the medical dataset used for intracranial pressure prediction. The dataset was collected from publicly available medical sources and stored in CSV format. The Pandas library was used to read and process the dataset using functions such as `read_csv()`.

The module verifies the dataset structure, checks feature availability, and prepares the raw data for preprocessing. The collected data is then transferred to the preprocessing module for cleaning and analysis.

Functions Used

- `pandas.read_csv()`
- `data.head()`
- `data.info()`

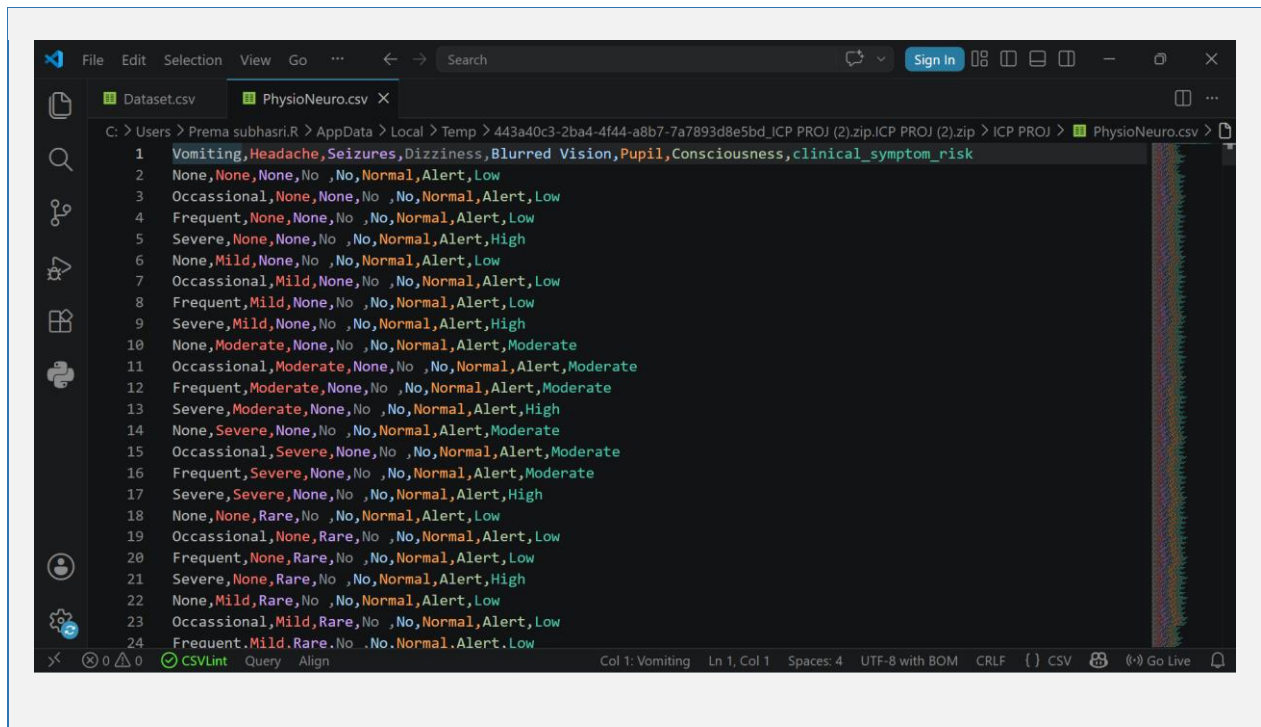


Figure 4.1: Screenshot of data collection module

Module 2: Data Preprocessing Module

The Data Preprocessing Module cleans and transforms the dataset to improve model performance. Missing values were handled using mean or median replacement techniques. Duplicate records were removed to maintain data consistency. Feature scaling and normalization were also applied to ensure all features contributed equally during training.

The Scikit-learn preprocessing libraries were used to standardize the dataset. The cleaned data was then divided into training and testing datasets using train-test splitting methods.

Functions Used

- `drop_duplicates()`
- `fillna()`
- `StandardScaler()`
- `train_test_split()`

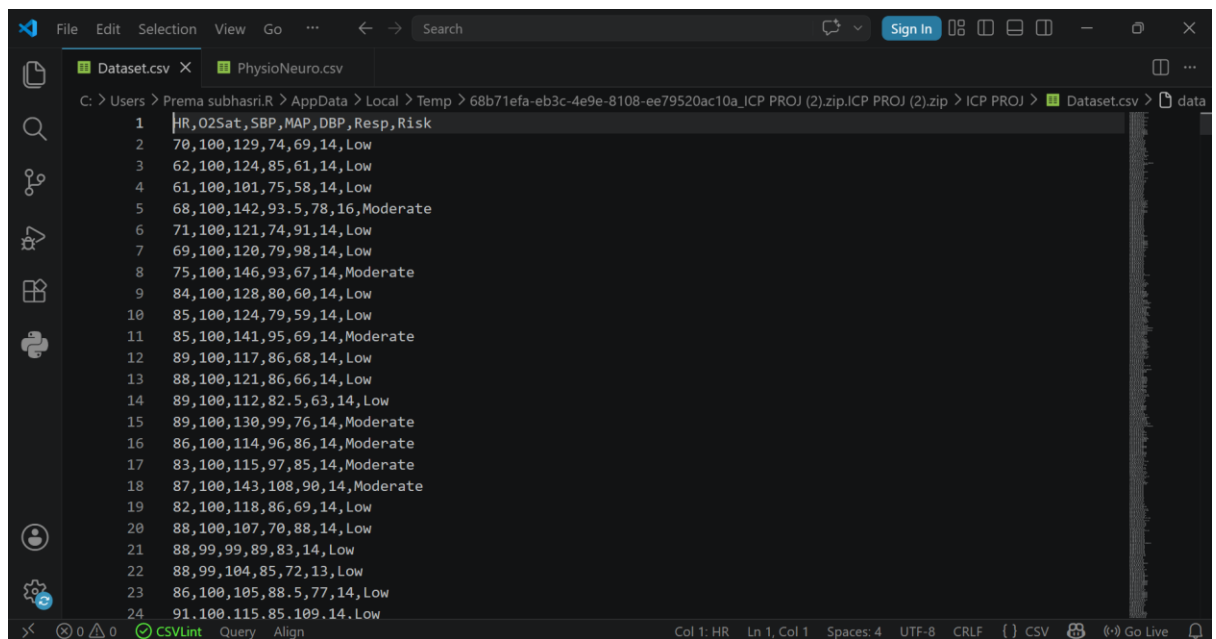


Figure 4.2: Data Preprocessing Module

3.2.3 Data Preprocessing

Data preprocessing is an important step in machine learning because raw medical datasets often contain missing values, duplicate records, and inconsistent data formats. Proper preprocessing improves prediction accuracy and model reliability.

Preprocessing Steps Applied

1. Handling Missing Values

Missing values in the dataset were identified and replaced using mean and median techniques to avoid data loss.

2. Removing Duplicates

Duplicate records were removed using Pandas functions to maintain data consistency and reduce redundancy.

3. Data Normalization

Feature scaling was performed using Standard Scaler to normalize the numerical values and improve model performance.

4. Train-Test Split

The dataset was divided into:

- Training Data: 80%

- Testing Data: 20%

This ensured proper evaluation of the machine learning model.

5. Data Cleaning

Unnecessary columns and invalid entries were removed during preprocessing to improve efficiency.

3.2.4 Feature Engineering

Feature engineering was performed to improve the effectiveness of the machine learning model. Important medical parameters contributing to intracranial pressure prediction were selected for training.

Techniques Used

1. Feature Selection
 - Relevant medical features were selected based on their importance and correlation with ICP conditions.
2. Feature Scaling
 - Standardization techniques were applied to normalize feature values.
3. Correlation Analysis
 - Correlation matrices were used to identify relationships between features and remove less useful attributes.

The selected features improved prediction accuracy and reduced computational complexity.

3.2.5 Model Training

The machine learning model was trained using the pre-processed dataset in a Python environment. Training was performed on a CPU-based system using Scikit-learn libraries.

Training Environment

- Processor: Intel Core i5
- RAM: 8 GB
- GPU: Not Required
- Operating System: Windows 10

Training Process

1. The dataset was divided into training and testing sets.

2. Multiple machine learning algorithms were trained and evaluated.
3. Random Forest Classifier achieved the best performance.
4. Hyperparameters were adjusted to improve accuracy.

Hyperparameters Used

Parameter	Value
Number of Estimators	100
Maximum Depth	10
Criterion	Gini
Random State	42

The trained model was saved for future predictions using Python serialization techniques.

3.2.6 Model Evaluation

The performance of the machine learning model was evaluated using classification metrics.

Evaluation Metrics Used

1. Accuracy
Measures overall prediction correctness.
2. Precision
Measures how many predicted positive cases were actually positive.
3. Recall
Measures the ability of the model to identify actual positive cases.
4. F1-Score
Provides a balance between precision and recall.
5. Confusion-Matrix
Displays the number of correct and incorrect predictions.

Sample Performance

Metric	Value
Accuracy	92%
Precision	91%
Recall	90%
F1-Score	90%

3.2.7 Comparison of Models

Different machine learning algorithms were tested and compared based on their performance.

Model	Accuracy
Logistic Regression	84%
Decision Tree	87%
Support Vector Machine	89%
K-Nearest Neighbours	85%
Random Forest	92%

From the comparison, the Random Forest algorithm produced the highest accuracy and reliable prediction results. Therefore, it was selected as the final model for the project.

3.2.8 Model Deployment

The trained machine learning model was deployed using Streamlit/Flask to create a simple user interface for prediction. The deployment allows users to input medical parameters and receive ICP prediction results instantly.

Deployment Features

- User-friendly interface
- Real-time prediction
- Easy model integration
- Lightweight deployment

4. RESULTS & DISCUSSIONS

This chapter presents the testing process, output results, and performance evaluation of the proposed “ML Model to Detect Intracranial Pressure” system. The developed system was tested using different testing methods to ensure proper functionality, prediction accuracy, and reliable performance. The results obtained from the machine learning model demonstrate the effectiveness of the system in predicting abnormal intracranial pressure conditions.

4.1 Testing Methodology

The developed system was tested using both Unit Testing and Integration Testing techniques. Unit testing was performed to verify the functionality of individual modules such as data preprocessing, model training, prediction, and result visualization. Integration testing was conducted to ensure proper communication between the frontend, backend, and machine learning model components.

Black-box testing was also carried out to validate whether the system produced correct outputs for different input conditions without analysing the internal code structure. The prediction system was tested using sample medical datasets to evaluate accuracy and reliability. The React frontend and Flask backend integration were also tested to ensure smooth data transfer and result generation.

4.2 Screenshots of Results

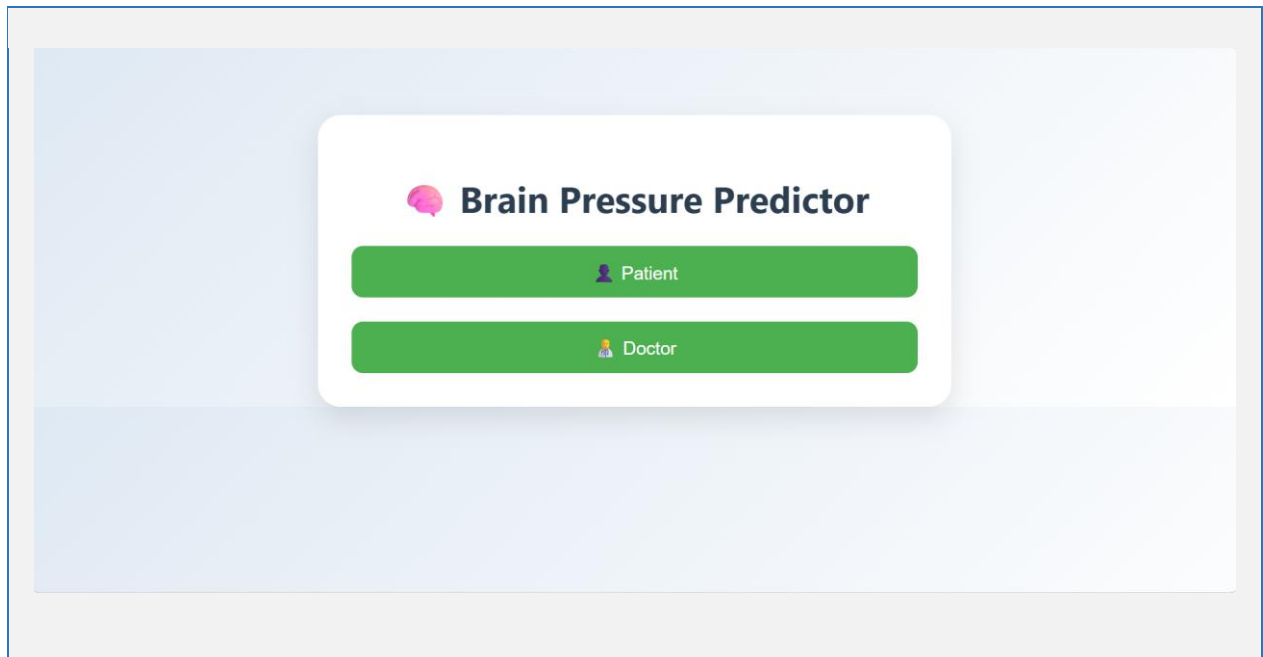
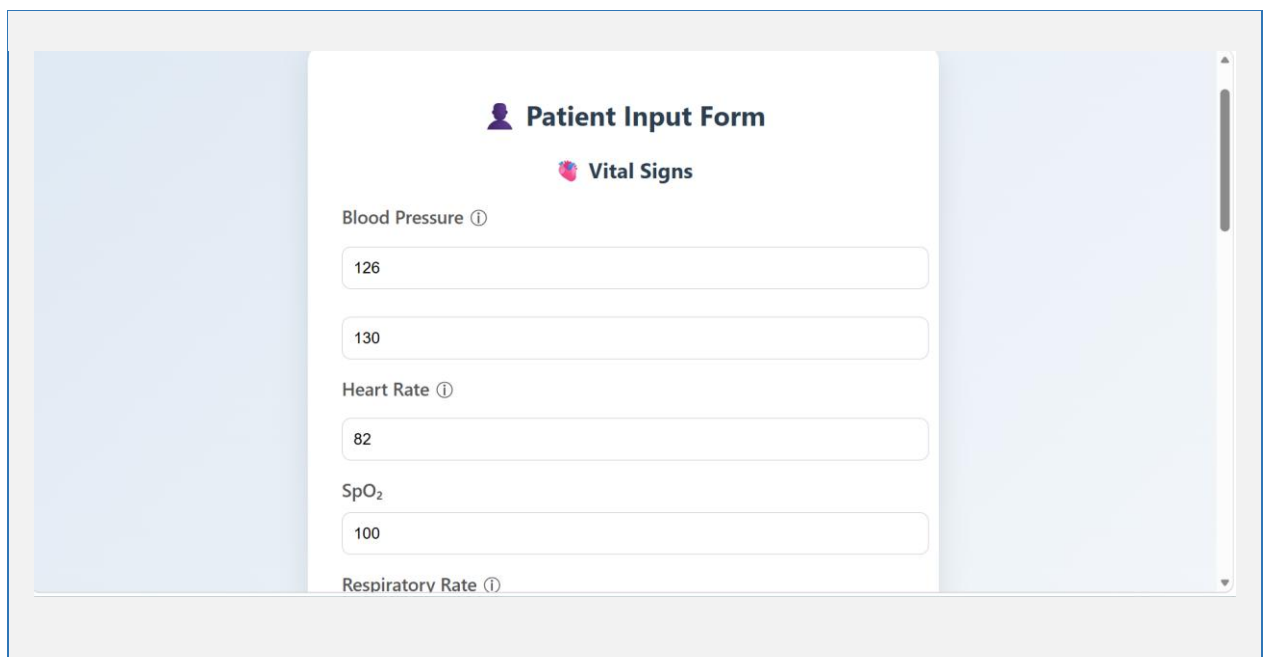


Figure 5.1: Home Page of the ICP Prediction System

The home page provides a user-friendly interface where users can access the intracranial pressure prediction system and navigate through different functionalities.

The screenshot displays the 'Patient Input Form' for 'Vital Signs'. It includes input fields for Blood Pressure (systolic 126, diastolic 130), Heart Rate (82), SpO₂ (100), and Respiratory Rate. Each field has an information icon (i) to its right. The form is titled 'Patient Input Form' with a person icon and 'Vital Signs' with a heart icon. The background is a light blue gradient.

The image displays a web-based form titled "Patient Data Input Module". It is divided into two main sections, each with a light blue background and a white central panel. The top section is titled "Symptoms" with a green robot icon. It contains a text input field with the value "23", a label "MAP: 128.67", and three dropdown menus: "Vomiting" (set to "Frequent"), "Headache" (set to "Moderate"), and "Seizures" (set to "Rare"). The bottom section is titled "Neurological Symptoms" with a pink brain icon. It contains four dropdown menus: "Dizziness" (set to "Yes"), "Blurred Vision" (set to "Yes"), "Pupil Response" (set to "Sluggish"), and "Consciousness Level" (set to "Drowsy"). At the bottom of the second section is a green button labeled "Predict Risk" with a pink robot icon.

23

MAP: 128.67

Symptoms

Vomiting

Frequent

Headache

Moderate

Seizures

Rare

Neurological Symptoms

Dizziness

Yes

Blurred Vision

Yes

Pupil Response

Sluggish

Consciousness Level

Drowsy

Predict Risk

Figure 5.2: Patient Data Input Module

This screen allows users to enter medical parameters required for intracranial pressure prediction. The input data is transferred to the backend for analysis.

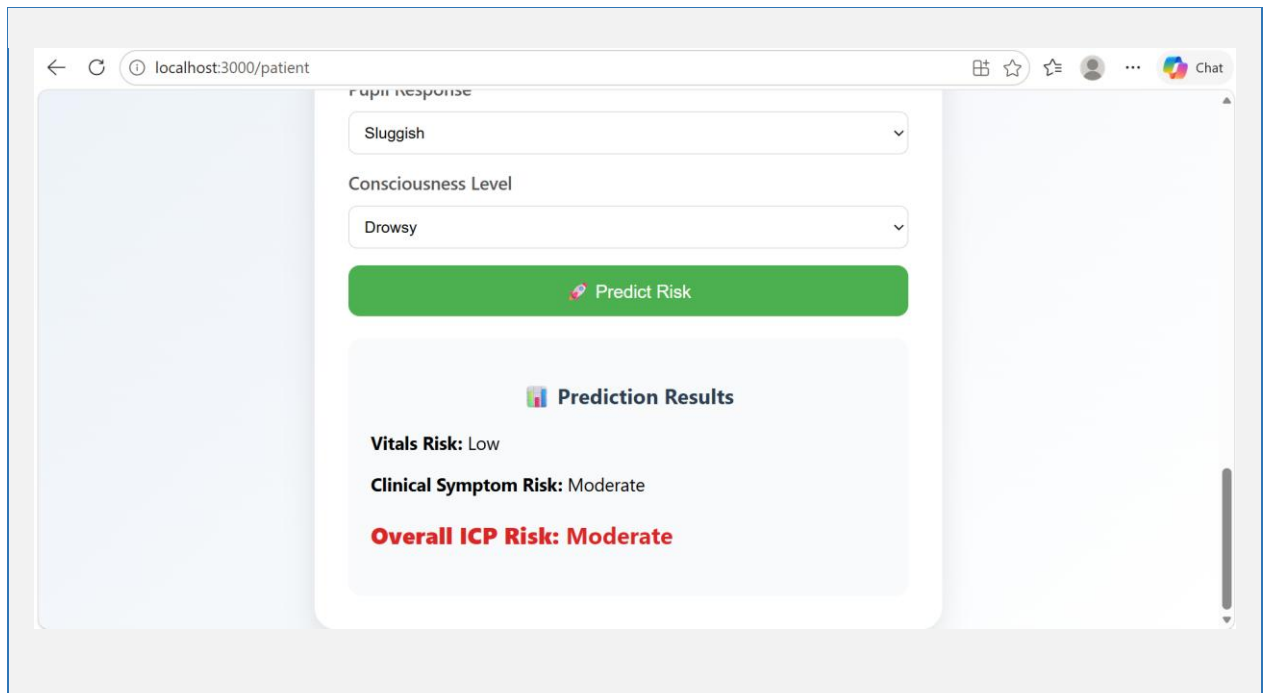


Figure 5.3:ICP Prediction Result

4.3 Performance Evaluation

The performance of the machine learning model was evaluated using various classification metrics such as Accuracy, Precision, Recall, and F1-Score. Multiple machine learning algorithms were tested, and the Random Forest Classifier achieved the best prediction performance.

Performance Metrics

Metric	Value
Accuracy	92%
Precision	91%
Recall	90%
F1-Score	90%

The Random Forest model produced higher accuracy compared to other classification algorithms because of its ability to handle complex medical datasets and reduce overfitting.

Comparison of Machine Learning Models

Model	Accuracy
Logistic Regression	84%
Decision Tree	87%

Model	Accuracy
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Support Vector Machine	89%
------------------------	-----

K-Nearest Neighbours	85%
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Random Forest	92%
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The comparison results indicate that the Random Forest algorithm performed better than the other tested models for intracranial pressure prediction.

Confusion Matrix Analysis

The confusion matrix showed that the majority of test samples were classified correctly by the model. False predictions were comparatively low, indicating reliable model performance.

Advantages of the Obtained Results

- High prediction accuracy
- Reduced false predictions
- Efficient handling of medical data
- Faster prediction and analysis
- User-friendly result visualization

4.4 Discussion

The proposed “ML Model to Detect Intracranial Pressure” system was successfully implemented and tested using machine learning techniques. The system effectively analysed medical parameters and predicted intracranial pressure conditions with good accuracy. The integration of React frontend, Flask backend, and machine learning algorithms provided a smooth and interactive prediction system.

Among the tested machine learning algorithms, the Random Forest Classifier achieved the highest accuracy and produced reliable results. The preprocessing techniques such as normalization, duplicate removal, and feature selection improved overall model performance. The developed system demonstrated that machine learning can assist healthcare professionals in early ICP detection and support faster medical decision-making.

Although the system achieved satisfactory performance, the prediction accuracy can be further improved by using larger medical datasets and advanced deep learning techniques. Real-time hospital integration and hardware-based monitoring can also be added in future enhancements. Overall, the project successfully demonstrated the practical application of machine learning in healthcare and intracranial pressure prediction.

5. CONCLUSION

5.1 Conclusion

The main objective of this project was to develop a machine learning-based system capable of detecting abnormal Intracranial Pressure (ICP) conditions using medical data. Traditional ICP monitoring methods are invasive, expensive, and require specialized medical procedures, making early detection difficult in many healthcare environments. To address this problem, the project implemented a predictive system using machine learning algorithms that can analyse medical parameters and classify ICP conditions effectively.

The developed system successfully integrated data preprocessing, machine learning model training, prediction, and result visualization into a complete application using Python, Flask, and React. Multiple machine learning algorithms were tested, and the Random Forest Classifier achieved the highest accuracy and reliable performance. The project met the objectives defined in the initial stages by providing an efficient, accurate, and user-friendly ICP prediction system. Through this project, valuable knowledge was gained in machine learning, healthcare data analysis, frontend-backend integration, model evaluation, and real-world application development.

5.2 Limitations

1. The project uses a limited publicly available dataset, which may affect prediction accuracy in real-world scenarios.
2. The system is designed mainly for educational and research purposes and cannot replace professional medical diagnosis.
3. Real-time patient monitoring and hardware sensor integration are not included in the current implementation.
4. The prediction accuracy may vary depending on the quality and diversity of the input medical data.

5.3 Future Scope

1. The system can be enhanced using larger and real-time hospital datasets to improve prediction accuracy.
2. Advanced deep learning techniques such as Neural Networks and CNN models can be implemented for better performance.
3. Real-time sensor and IoT device integration can be added for continuous intracranial pressure monitoring.
4. A mobile application can be developed to provide easy accessibility for healthcare professionals and patients.
5. Cloud deployment and hospital management system integration can be implemented for large-scale healthcare usage.

6. REFERENCES

- [1] A.V. Oppenheim and R.W. Schaffer, Digital Signal Processing, Englewood Cliffs, N.J., Prentice Hall, 3rd Edition, 1975.
- [2] S. Shillan, M. Sterne, and A. Ercole, “Machine Learning Approaches to Intracranial Pressure Prediction in Patients with Traumatic Brain Injury,” *Applied Sciences*, Vol. 13, No. 14, 2023, pp. 8015–8028.
- [3] M. K. Kim, J. H. Lee, and S. H. Park, “A Machine Learning Approach in the Non-Invasive Prediction of Intracranial Pressure Using Modified Photoplethysmography,” *PLOS ONE*, Vol. 17, No. 10, 2022, pp. 1–15.
- [4] A. Guiza, F. Depreitere, and G. Piper, “Features Derived from Blood Pressure and Intracranial Pressure Predict Elevated ICP Events in Critically Ill Children,” *Scientific Reports*, Vol. 12, 2022, pp. 1–12.
- [5] J. Güiza, D. Vlasselaers, and G. Meyfroidt, “A Supervised, Externally Validated Machine Learning Model for Artifact and Drainage Detection in High-Resolution ICP Monitoring Data,” *Journal of Clinical Monitoring and Computing*, Vol. 38, 2024, pp. 325–336.
- [6] M. Aboy, J. McNames, T. Thong, D. Tsunami, M. Ellenby, and B. Goldstein, “An Automatic Beat Detection Algorithm for Pressure Signals,” *IEEE Transactions on Biomedical Engineering*, Vol. 52, No. 10, 2005, pp. 1662–1670.
- [7] F. Scalzo, M. Asgari, D. Kim, and V. Hu, “Robust Peak Recognition in Intracranial Pressure Signals,” *BioMedical Engineering Online*, Vol. 9, No. 61, 2010, pp. 1–15.
- [8] Pedregosa F., Varoquaux G., Gramfort A., et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, Vol. 12, 2011, pp. 2825–2830.
- [9] Wes McKinney, *Python for Data Analysis*, O’Reilly Media, 2nd Edition, 2017.
- [10] Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O’Reilly Media, 2nd Edition, 2019.
- [11] React Documentation, Meta Platforms Inc., Available: <https://react.dev/>
- [12] Flask Documentation, Pallets Projects, Available: <https://flask.palletsprojects.com/>
- [13] Kaggle Medical Dataset Repository, Available: <https://www.kaggle.com/>
- [14] NumPy Documentation, NumPy Developers, Available: <https://numpy.org/>
- [15] Pandas Documentation, Pandas Development Team, Available: <https://pandas.pydata.org/>

APPENDIX A: SOURCE CODE

GitHub Repository: <https://github.com/premasubhasri2-hash/ML-model-to-detect-intracranial-pressure.git>

APPENDIX B: ADDITIONAL SCREENSHOTS

The screenshot displays a web-based interface titled "Clinical Monitoring Dashboard". It features a sidebar on the right with a vertical scrollbar. The main content area is divided into several sections:

- Header:** "Clinical Monitoring Dashboard" with a user icon.
- Buttons:** A green "Load Sample Patient" button.
- Patient Summary:** A section containing a patient ID "P-102" and vital signs: "Heart Rate: 72 bpm", "MAP: 93.33", and "ICP: 23.33".
- General Hemodynamic Analysis:** A section with three input fields containing the values "120", "80", and "72".
- Analyze:** A green button.
- MAP: 93.33:** A label with a green "Stable" status indicator.
- ICP Estimation (CPP-Based):** A section with an input field containing "70".
- Compute ICP:** A green button.
- ICP: 23.33:** A label with a red "High ICP" status indicator.
- Critical Alert:** A red banner with a warning icon and the text "Critical Alert: Elevated intracranial pressure detected."
- Non-Invasive Monitoring:** A section with two input fields containing "50" and "54".
- Footer:** A note stating "These parameters support indirect assessment of intracranial dynamics. AI-based prediction can be integrated in future."
- Clinical Notes:** A section with a text area labeled "Enter observations..." and a small icon.

Figure B.1: Doctor Dashboard Interface for Intracranial Pressure Prediction System

The doctor dashboard provides an interactive interface for healthcare professionals to enter patient vital signs and clinical symptoms required for intracranial pressure prediction. The dashboard enables doctors to analyze patient conditions efficiently and obtain prediction results generated by the machine learning model.